

AADYA SHARMA

Lucknow, India

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RESEARCH INTERESTS

Gravitational lensing, modified gravity, black holes, lensed blazars, observational data fitting

EDUCATION

Course	Institute [India]	CGPA / Grade	Passing Year
PhD in Relativistic Astrophysics	GLA University, Mathura, Uttar Pradesh	8.71	Ongoing
MSc Physics	Banasthali Vidyapith, Rajasthan	7.50	2024
Advanced Diploma in German Language	Banasthali Vidyapith, Rajasthan	Pass	2023
BSc [Mathematics, Physics, CS]	Banasthali Vidyapith, Rajasthan	9.23	2022
12 th CBSE	Bal Vidya Mandir Sr. Sec. School, Lucknow	8.33	2018
10 th CBSE	Bal Vidya Mandir Sr. Sec. School, Lucknow	10	2016

EXPERIENCE

GLA University, Mathura	Junior Research Assistant in UP-CST project entitled “Particle Motion, Gravitational Lensing, and accretion disk around the black holes in modified gravity constrained by observational data”	Oct 2024 - Present
IUCAA, Pune	Summer Student for Astronomy and Astrophysics summer school	May 2022 - Jun 2022
Navjeevan Publication, Jaipur	German to Hindi translator for ‘Es war einmal... Select Fairy Tales of the Grimm Brothers’	Aug 2022 - Nov2022
Spaceonova, Delhi	Astrobiology Ambassador to enhance astrobiology outreach among students	Oct 2021 - Jan 2022

PROJECTS AND MODELS

- X-ray analysis of astrophysical binary systems GX339-4 and 1RXS J121902.7-630910* | Duration: 10 months - Performed Xray analysis of GX339-4 and 1RXS J121902.7-630910 using XSPEC software, identifying black hole signatures in GX339-4 and suggesting a potential white dwarf association for 1RXS J121902.7-630910.
- Desktop Voice Assistant using Python* | Duration: 4 months - Designed and developed a desktop voice assistant using Python, enabling voice-based command execution, task automation, and real-time interaction.
- Working model presentation of Hologram* - Developed and presented a working model of holography, demonstrating the creation of 3D visual representations using light diffraction
- Working model presentation of LiFi: data transmission using light* - presented a working model of LiFi technology, showcasing high-speed data transmission using visible light for wireless communication.

TECHNICAL SKILLS

Python, Mathematica, C, C++, MATLAB, XSPEC, LaTeX Overleaf, Linux OS, SQL, Actuarial Science

CERTIFICATES

Conferences:

- International Conference on Astroparticle Physics, Gravitation and Cosmology [APGC 2025] – GLA University
- National Conference on Preserving and Promoting Indian Intellectual Heritage – K.L.S. College, Nawada

Workshops:

- Theoretical Aspects of Relativistic Jets – IUCAA
- Astrophysical Dust Ices - Physical Research Laboratory
- Rocket & Satellite Engineering - International Academy of Space Law
- Python with Space Applications - Spaceonova

Courses:

- Python Programming - Banasthali Vidyapith
- Masterclass on Astrobiology - Spaceonova
- Actuarial Sciences – Banasthali Vidyapith
- Women’s Business Creator - Arizona University

EXTRACURRICULAR

Organiser/Coordinator: National Science Day 24, Mayukh Tech Fest 24 & 23, Literati Fest 23, TDDS 23, and Dimension 22

Roles: Hostel Representative in 2022-23, and Class Representative in 2020-22.

Activities: Radio Anchoring, Extension Program for Women Empowerment, Creative Arts, and Martial Arts.

Hobbies: Reading, writing poetry, nature photography, stargazing, painting, DIY, chess, cycling, gardening

LANGUAGES

Hindi: C2, English: B2, German: A1